



University of Dhaka
Department of Computer Science and Engineering

SOFTWARE REQUIREMENTS SPECIFICATION

CSEDU Nexus

CSEDU Students' Club (CSEDUSC)

A Centralized Digital Platform for Club Administration

CSE 4113- Internet Programming Lab

Submitted By

Tawyabul Islam Tamim — Roll: 04

Md. Abu Horaira — Roll: 26

Mst. Irin Sultana - Roll: 41

Md. Adib Ahsan — Roll: 60

Submitted To

Dr. Md. Mamun-Or-Rashid , Professor, CSE, University of Dhaka

Md. Fahim Arefin, Lecturer, CSE, University of Dhaka

Md. Ahasanul Alams, Lecturer, CSE, University of Dhaka

March 2026

0 Contents

1	Document Control	3
1.1	Metadata	3
1.2	Authors and Reviewers	3
1.3	Revision History	3
2	Problem Definition	4
2.1	Problem Statement	4
2.2	Target Users	5
2.3	Goals and Non-Goals	5
3	Scope and Context	6
3.1	In Scope	6
3.2	Out of Scope	6
3.3	Assumptions and Constraints	6
4	Stakeholders and Roles	8
4.1	Stakeholders	8
4.2	User Roles and Permissions	9
5	Functional Requirements	10
6	User Stories and Use Cases	14
7	Data Requirements	16
7.1	Core Entities	16
7.2	Data Dictionary	18
7.3	Data Model	19
7.3.1	Entity Relationships and Business Rules	19
8	UI and UX Requirements	22
8.1	Wireframes	22
8.2	UX and Accessibility Baseline	22
9	Non-Functional Requirements	23
10	Risks and Open Questions	24
10.1	Risk Register	24

10.2 Open Questions	25
11 Testability and Traceability	26
11.1 Requirement Traceability	26
11.2 Initial Manual Test Plan	27
12 Lab 2 Submission Checklist	29
A Entity-Relationship Diagram	30
B UML Diagram	31
C Wireframes	32

1 Document Control

1.1 Metadata

Team	Ravenclaw
Project Name	CSEDU Nexus
Repository URL	https://github.com/puHoraira/CSEDU_Nexus
Version	0.1
Status	Draft
Date	06 March 2026

1.2 Authors and Reviewers

Role	Name	ID / Email	Date
Author	Md. Abu Horaira	mdabu-2021911202@cs.du.ac.bd	March 2026
Author	Tawyabul Islam Tamim	tawyabulislam-2021411180@cse.du.ac.bd	March 2026
Team Reviewer	Md. Adib Ahsan	mdadib-2021211236@cs.du.ac.bd	March 2026
Team Reviewer	Mst. Irin Sultana	mstirin-2021311217@cs.du.ac.bd	March 2026

1.3 Revision History

Version	Date	Author	Changes
0.1	March 2026	- Tawyabul Islam - Md. Abu Horaira	Initial draft based on CSEDUSC Constitution

2 Problem Definition

2.1 Problem Statement

The CSEDU Students' Club (CSEDUSC), formally constituted under the Department of Computer Science and Engineering, University of Dhaka, currently lacks a centralised digital platform to manage its members, events, elections, finances, and committee structures. Key pain points include:

- Manual, paper-based membership registration and tracking.
- No systematic mechanism to enforce eligibility rules for elections (e.g., year of study, prior EC experience).
- Absence of a transparent financial ledger accessible to moderators and the chief patron.
- Difficulty tracking executive committee attendance, meeting minutes, and quarterly expenditure reports.
- No structured process for certificate of membership requests.

Without a unified system, the club risks constitutional violations, reduced transparency, and administrative overhead – ultimately hampering its mission to foster students' holistic growth through co-curricular and extra-curricular activities.

The absence of a centralised platform creates systemic inefficiencies. This SRS defines the requirements for a digital solution that enforces the CSEDUSC Constitution programmatically, ensuring transparency, accountability, and constitutional compliance across all club operations.

2.2 Target Users

User Group	Description	Main Needs
General Members	All current CSEDU undergraduate students	Register/view membership, view events, sign-up as volunteers
EC Members	Elected posts 1–11 and executive members	Manage events, budgets, meetings, committees
Moderator(s)	Current Student Advisor(s) of the department	Oversight, approval workflows, financial audit
Chief Patron	Department Chairman	Final approval, overrule decisions, abolish EC
Election Commission	Three-member commission	Conduct and manage the two-phase election
Alumni	Former CSEDU students	View events, act as election commissioners

2.3 Goals and Non-Goals

GOALS

- Digitise membership registration and lifecycle management (including cancellation rules per Articles VI–VII).
- Automate eligibility checks for election candidacy per Article XIV and XV.
- Provide a transparent financial management module aligned with Article XII.
- Enable online executive meeting scheduling, attendance recording, and minute-keeping (Article XVI).
- Support an Annual General Meeting (AGM) workflow and election commission formation (Article XVIII).
- Provide certificate-of-membership request and issuance workflow (Article XIX).

NON-GOALS

- Integration with the university’s academic result system.
- Full Learning Management System (LMS) or course delivery capabilities.
- Mobile-native (iOS/Android) application in v1.
- Automated financial transactions / online payment gateway in v1.

3 Scope and Context

3.1 In Scope

- Member registration, profile management and membership-status tracking.
- EC formation, role assignment, impeachment, resignation and vacancy-filling workflows.
- Two-phase election module with eligibility validation.
- Event planning, volunteer management, and activity calendar aligned with Article XIII mandatory activities.
- Financial module: budget submission, fund tracking, quarterly expenditure reports and audit trail.
- Meeting scheduler, attendance recording.
- Constitution change proposal and approval workflow (Article XVII).
- AGM workflow and election commission formation.
- Certificate-of-membership request and issuance.
- Role-based access control (RBAC) for all user groups.
- Notification system (email) for meetings, elections, and notices.
- Online payment system for events and various payment.

3.2 Out of Scope

- Integration with university ERP or student information systems.
- Native mobile applications.
- Social-media auto-posting (only link sharing).

3.3 Assumptions and Constraints

ASSUMPTIONS

- All members and executives have access to internet-enabled devices (desktop, laptop, or tablet) with modern web browsers.
- Users possess basic computer literacy and familiarity with web-based applications.
- The CSEDUSC Constitution remains stable during development; any amendments will be incorporated in subsequent releases.
- The university administration supports digital transformation of club operations and grants necessary permissions for system deployment.
- All financial records and member data will be maintained digitally with physical backups where legally required.

TECHNICAL CONSTRAINTS

- *Platform Limitation:* The system must be delivered as a responsive web application accessible through standard web browsers; native mobile applications are excluded from the

initial scope.

- *Infrastructure*: University internet bandwidth may fluctuate (especially during peak hours); the application must remain functional with network low speed.
- *Performance*: The system must handle at least 500 concurrent users during peak activity periods (e.g., election voting, event registration).
- *Compatibility*: Must support desktop and tablet viewports; mobile optimization is secondary priority for v1.

ORGANISATIONAL CONSTRAINTS

- System must strictly enforce rules specified in the CSEDUSC Constitution.
- Chief Patron and Moderator must retain override/approval capabilities at all times.
- Membership cancellation notices require multi-party digital signatures.

TIMELINE CONSTRAINTS

- *Lab 2 (Current)*: SRS v1, problem definition, user stories, wireframes, and initial data model must be completed and submitted for peer evaluation.
- *Lab 3 (Peer Review)*: SRS revision based on peer feedback must be completed before sprint development begins.
- *Lab 4 (Sprint 1 – MVP)*: Working vertical slices delivering end-to-end functionality for at least three core modules (membership registration, basic event listing, and user authentication). SDD documentation and public repository link submission required. Target features:
 - Member registration and login system
 - Role-based access control (RBAC) foundation
 - Basic event creation and viewing
 - Public portal homepage
- *Lab 5 (Sprint 2 – Core Expansion)*: Complete remaining core workflows with improved reliability and error handling. Automatic deployment pipeline operational. Target features:
 - Executive committee dashboard
 - Event volunteer management
 - Meeting scheduler and attendance tracking
 - Financial module foundation (budget submission)
 - Election management
- *Lab 6 (UI Review)*: System must be deployed and accessible for peer usability/accessibility critique after two sprints of development.
- *Lab 7 (Testing Baseline)*: Unit, integration, and end-to-end test suite operational.
- *Lab 8 (Peer QA)*: Adversarial QA testing by peer teams; all critical bugs from Lab 6 UI review resolved.
- *Lab 9 (Hardening)*: Address critical and high-priority bugs from peer QA; finalize documentation, deployment runbook, and production readiness checklist.
- *Lab 10 (Final Defense)*: Fully functional system with live demo capability, comprehensive test evidence, final report, video demonstration, and complete GitHub repository with deployment instructions.
- *External Constraint*: MVP (Labs 4–5) should align with CSEDUSC operations. pilot deployment with 10–15 executive members during Lab 9 for real-world validation.

4 Stakeholders and Roles

4.1 Stakeholders

Stakeholder	Interest	Priority
Chief Patron (Chairman)	Constitutional compliance, final approvals	High
Moderator(s)	Day-to-day oversight, financial audit	High
EC Members	Efficient event and club management	High
General Members	Access to events, membership benefits	Medium
Election Commission	Fair and transparent elections	High
Alumni	Advisory roles, event participation	Low
Department Admin Office	Financial document submission	Medium

4.2 User Roles and Permissions

Role	Description	Permissions
General Member	Any current CSEDU student	View events, register, sign membership form, apply as volunteer
President	Post-1 EC, 3rd year+, 2+ years EC experience	Preside over meetings, call meetings, convene constitution-change committee
Vice President	Post-2 EC, 3rd year+, 2+ years EC experience	Support President, act as President in absence, oversee planned activities
General Secretary	Post-3 EC, 3rd year+, 2+ years EC experience	Call meetings, organize/monitor ad-hoc committees, manage volunteers during events
AGS (Organization)	Post-4 EC, 3rd year+, 1+ year EC experience	Organize volunteers during programs, assist GS in duties
AGS (Public Relations)	Post-5 EC, 3rd year+, 1+ year EC experience	Contact alumni/media/clubs, send invitations, maintain website, manage online events
Treasurer	Post-6 EC, 3rd year+, 1+ year EC experience	Maintain fund records, manage collection/expenditure, submit quarterly reports
Secretary (Publication)	Post-7 EC, 2nd year+	Print banners/posters/notices, produce annual reports/magazines
Secretary (Sports)	Post-8 EC, 2nd year+	Maintain games equipment, organize tournaments, coordinate official teams
Secretary (Seminars)	Post-9 EC, 2nd year+	Arrange mandatory higher study seminar, organize workshops/training/invited talks
Secretary (Cultural)	Post-10 EC, 2nd year+	Arrange formal/informal cultural sessions throughout the year
Secretary (Graphics)	Post-11 EC, 2nd year+	Design and curate content for publication and social media
Executive Members	Posts 12-onward EC	Assist Posts 1-11, lead volunteer groups during events (2nd year+)
Chief Patron	Department Chairman	Full override, approve constitution changes, issue/cancel memberships, sign cheques
Moderator	Student Advisor(s)	Approve proposals, view financial records, manage election commission, candidacy cancel
Election Commissioner	Three-member commission	Manage election phases, validate candidacy, publish results
Alumni	Former CSEDU students	View events, election commission (eligible)

5 Functional Requirements

Table 1: Functional Requirements

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-CSC-001	Membership Registration	System shall allow any current CSEDU student to register as a general member by completing a digital membership form and signing the checklist.	M	Member appears in database with status “Active” after form submission and confirmation.
FR-CSC-002	Membership Cancellation	System shall automatically cancel membership upon expiration of studentship (final result published).	M	Membership status changes to “Expired” within 24 hours of result publication trigger.
FR-CSC-003	Cancellation Notice	System shall generate a cancellation notice signed digitally by Chief Patron, Moderator(s), and President upon manual cancellation.	M	Notice PDF is generated with all three digital signatures and published to the notice board.
FR-CSC-004	EC Formation	System shall support creation of an Executive Committee with roles as defined in Article V.	M	All 11 named posts plus executive members can be assigned; roles are stored with batch and year information.
FR-CSC-005	Election Phase 1	– System shall allow batch-level voting to select five representatives per batch.	M	Vote tallies are recorded; top 5 per batch are automatically marked as EC members.
FR-CSC-006	Election Phase 2	– System shall allow all undergraduate students to vote for Posts 1–11 from Phase-1 representatives who meet eligibility criteria.	M	Eligibility check runs automatically; ineligible candidates are blocked from appearing on ballot.
FR-CSC-007	Candidacy Eligibility	System shall enforce year-of-study and EC-experience requirements per Article XIV and XV for each post.	M	Attempt to add ineligible candidate returns error with specific rule violated.

Continued on next page...

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-CSC-008	Impeachment Workflow	System shall require a minimum two-thirds vote of total EC members to accept impeachment.	M	Impeachment is recorded only when $\geq 2/3$ of EC members vote in favour; notice is auto-generated.
FR-CSC-009	Resignation Workflow	System shall allow President to resign directly to Moderator, and other EC members to resign via President to Moderator.	M	Resignation letter is submitted digitally; post status changes to “Vacant” upon Moderator acknowledgement.
FR-CSC-010	Vacancy Filling	System shall trigger a voting process from the corresponding batch when any EC post becomes vacant.	M	Vacancy notice is sent to eligible batch members; vote results fill the post.
FR-CSC-011	Meeting Scheduling	System shall allow President or General Secretary to schedule executive meetings with at least one week’s notice.	M	Meeting notice appears on dashboard and is sent via notification to all EC members.
FR-CSC-012	Meeting Attendance	System shall record digital attendance with signature confirmation for each executive meeting.	M	Attendance sheet with member names and digital sign-off is stored; sheet is shareable with Moderator.
FR-CSC-013	Meeting Minutes	System shall allow General Secretary to record and store meeting minutes linked to each meeting.	M	Minutes are stored, versioned, and accessible to EC members and Moderator.
FR-CSC-014	Absence Tracking	System shall automatically flag an EC member who has been absent from three consecutive meetings without notice.	M	System sends alert to President and Moderator after third consecutive unexcused absence.
FR-CSC-015	Event Management	System shall allow EC members to create, update, and delete events with associated budgets and volunteer requirements.	M	Event is visible on the calendar; budget request is routed to Moderator for approval.

Continued on next page...

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-CSC-016	Mandatory Activity Checklist	System shall track and display completion status of mandatory annual activities listed in Article XIII.	S	Dashboard shows completion percentage of mandatory activities for the current EC year.
FR-CSC-017	Volunteer Management	System shall allow general members to register as volunteers for events and allow EC members to assign volunteer roles.	M	Volunteer list per event is maintained; assignment confirmation is sent to volunteers.
FR-CSC-018	Budget Submission	System shall allow newly formed EC to submit proposed expenditure and expected revenues within one month of formation.	M	Budget submission is timestamped; late submission triggers a Moderator alert.
FR-CSC-019	Expenditure Report	System shall allow Treasurer to prepare and submit a quarterly expenditure report signed digitally by Treasurer and Moderator.	M	Report is generated from transaction records, signed, and stored in the department office section.
FR-CSC-020	Fund Tracking	System shall maintain a running ledger of all fund inflows and outflows.	M	Ledger entries include date, amount, category, and reference document; balance is always visible to Moderator and Chief Patron.
FR-CSC-021	Constitution Change	System shall allow President to initiate a constitution change proposal, collect EC two-thirds vote, and route to Moderator and Chief Patron for approval.	S	Proposal workflow enforces one-week notice; change is published only after Academic Committee approval is recorded.
FR-CSC-022	AGM Workflow	System shall support the Annual General Meeting process including notice, annual report submission, financial ledger presentation, and election commission formation.	M	AGM checklist is completed within the system; all submissions are archived.

Continued on next page...

Req ID	Feature	Requirement Statement	Priority	Acceptance Criteria
FR-CSC-023	Certificate of Membership	System shall allow EC members to request a certificate of membership issued digitally by the Chief Patron.	S	Certificate is generated from system records, approved by Chief Patron, and downloadable as PDF.
FR-CSC-024	Role-Based Access Control	System shall enforce role-based permissions so that each user can only access features relevant to their role.	M	Unauthorised access attempts return a 403 error; permission matrix matches Section 4.2.
FR-CSC-025	Notification System	System shall send in-app and email notifications for meetings, election announcements, membership status changes, and budget approvals.	M	Notification is delivered within 5 minutes of the triggering event; user can configure email opt-out.

6 User Stories and Use Cases

Table 2: User Stories

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-CSC-001	General Member	to register online with a membership form and checklist	I can officially become a CSEDUSC member	Form submitted; confirmation email received; member appears in active list.
US-CSC-002	General Member	to view all upcoming club events and activities	I can plan my participation	Events page loads with title, date, venue, and registration link.
US-CSC-003	General Member	to sign up as a volunteer for an event	I can contribute to club activities	Volunteer registration recorded; confirmation shown on dashboard.
US-CSC-004	EC Member (General Secretary)	to schedule an executive meeting with automatic notifications	all EC members are informed in time	Meeting created; notifications sent to all EC members immediately.
US-CSC-005	EC Member (General Secretary)	to record digital attendance for each executive meeting	absence tracking is accurate and automatic	Attendance sheet stored; members absent for 3 consecutive meetings are flagged.
US-CSC-006	EC Member (Treasurer)	to log all income and expenditure entries	the financial ledger is always up to date	Entry saved with timestamp, category, and amount; balance updates immediately.
US-CSC-007	EC Member (Treasurer)	to generate and submit a quarterly expenditure report	I can meet the constitutional reporting requirement	Report generated from ledger; digital signature workflow initiated for Moderator sign-off.

Continued on next page...

Story ID	As a...	I want...	So that...	Acceptance Criteria
US-CSC-008	EC Member (President)	to initiate an impeachment vote for an EC member	constitutional procedures are followed	Vote created; $\geq 2/3$ acceptance triggers auto-generated notice; membership cancelled.
US-CSC-009	EC Member	to submit my resignation letter digitally through the President	the process is documented and transparent	Resignation routed to Moderator; post marked "Vacant" upon acknowledgement.
US-CSC-010	Moderator	to review and approve all budget submissions and expenditure reports	financial transparency is maintained	Approval or rejection recorded with comment; Treasurer notified immediately.
US-CSC-011	Moderator	to oversee the two-phase election process	the election is conducted fairly and constitutionally	Election phases are visible to Moderator; any anomaly can be flagged before results are published.
US-CSC-012	Chief Patron	to digitally approve or overrule any EC decision	my authority as defined in the constitution is exercised	Approval/overrule action is logged with timestamp and reason; all EC members are notified.
US-CSC-013	Election Commissioner	to validate candidacy eligibility automatically	only eligible candidates appear on the ballot	Ineligible candidates are blocked with a reason message; eligible list is published to Moderator.
US-CSC-014	EC Member	to request a certificate of membership listing my contributions	I can use it for academic or professional purposes	Request submitted; Chief Patron approves; PDF certificate generated and downloadable.
US-CSC-015	EC Member (President)	to initiate a constitution-change proposal and collect votes	amendments follow the process in Article XVII	Proposal created; one-week notice sent; $\geq 2/3$ vote recorded; routed to Moderator and Chief Patron.

7 Data Requirements

7.1 Core Entities

Table 3: Core Entities of CSEDUSC System

Entity	Description	Key Fields	Constraints
Member	Any CSEDU student registered in the club	memberID, name, studentID, batch, status, joinDate, email, phone	Status $\in$ {Active, Cancelled, Expired}
Alumni	Former CSEDU students	alumniID, memberID, graduationYear, email, phone, isEligibleCommissioner	Can serve on election commission
ECMember	Elected executive committee member	ecMemberID, memberID, post, term, batch, ECExperience, startDate, endDate	One active EC per post per term
Election	Full election cycle	electionID, phase, startDate, endDate, status, commissionID	Two phases required; Status: Draft/Active/Completed
Election Commission	Three-member commission	commissionID, electionID, chairperson, member1, member2, formationDate	Must have exactly 3 members; at least 2 must be alumni
Vote	Single vote in election	voteID, voterID, candidateID, electionID, phase, timestamp, isAnonymous	One vote per voter per phase; immutable
Event	Club event or activity	eventID, title, type, venue, budget, status, organizerID	Type $\in$ {Mandatory, Optional}; Status $\in$ {Planned, Ongoing, Completed, Cancelled}
Volunteer	Event volunteer assignment	volunteerID, memberID, eventID, role, assignedBy, status	Status $\in$ {Registered, Assigned, Completed}

Continued on next page

Entity	Description	Key Fields	Constraints
Meeting	EC meeting	meetingID, date, time, calledBy, venue, minutes, quorum, status	Minimum one per two months; quorum $\geq 2/3$ EC
Meeting Attendance	Attendance record	attendanceID, meetingID, memberID, isPresent, digitalSignature, timestamp	Only for EC members; tracks absences
Transaction	Financial ledger entry	transactionID, type, amount, date, category, reference, createdBy, approvedBy	Type $\in \{\text{Income, Expenditure}\}$; immutable
Expenditure Report	Quarterly report	reportID, quarter, year, totalIncome, totalExpense, netBalance, treasurerSign, moderatorSign	Generated every 3 months
Certificate	Membership certificate	certID, memberID, requestDate, approvalDate, approvedBy, pdfURL, status	Issued by Chief Patron only
Constitution Change	Amendment proposal	changeID, proposedBy, description, proposalDate, voteCount, status, approvedBy	Once per EC year; needs $\geq 2/3$ EC + Moderator + Chief Patron approval
Moderator	Student Advisor	moderatorID, name, email, department, assignedDate	Oversight role; approves budgets and reports
Chief Patron	Department Chairman	patronID, name, email, department, appointmentDate	Final authority; signs cheques and certificates

7.2 Data Dictionary

Field	Type	Required	Validation / Constraint	Example
memberID	UUID	Yes	Auto-generated, unique	mem-0042
studentID	String	Yes	University student ID format	2021-045-343
batch	Integer	Yes	Valid CSEDU batch year	2021
status	Enum	Yes	Active, Cancelled, Expired	Active
post	String	No	Must be one of the 11 named posts	General Secretary
ECExperience	Integer	Yes	Years of experience in EC ≥ 0	2
amount	Decimal	Yes	Must be > 0 ; max 2 decimal places	5000.00
category	Enum	Yes	Dept / Donation Sponsor / RegFee / Expenditure	Donation
phase	Integer	Yes	1 or 2	1
quorum	Boolean	Yes	$\geq 2/3$ EC must be present	true

7.3 Data Model

The system's data architecture consists of sixteen core entities with well-defined relationships governing membership, alumni engagement, governance, events, elections, and financial operations. This model ensures constitutional compliance and data integrity across all workflows.

Entity Relationships and Business Rules

Table 4: Entity Relationships and Business Rules

Relationship	Cardinality	Description	Business Rules
Member → Alumni	1:0..1	A Member becomes an Alumnus upon graduation	Automatic transition when studentship expires
Member → ECMember	1:0..1	A Member may become an EC member	One active EC role per member per term; historical EC records retained for a specific term
Member → Vote	1:N	A Member casts votes in elections	One vote per member per election phase; votes are immutable
Member → Volunteer	1:N	A Member volunteers for events	Members sign up; EC assigns specific roles
Member → MeetingAttendance	1:N	EC Members have attendance records	Tracks presence/absence with digital signature; auto-flags 3 consecutive absences
Member → Certificate	1:N	A Member can request multiple certificates	Each request requires Chief Patron approval; one active request at a time
Alumni → Election Commission	1:N	Alumni can serve on election commission	At least 2 of 3 commission members must be alumni
ECMember → Event	1:N	An EC member organizes events	Each event assigned to one responsible EC post (e.g., Secretary Cultural)
ECMember → Transaction	1:N	Treasurer logs transactions	Only Treasurer role can create/edit transactions; requires supporting documents
ECMember → Meeting	1:N	EC members call and attend meetings	President or General Secretary can call meetings
ECMember → ConstitutionChange	1:N	EC members propose amendments	President initiates; needs $\geq 2/3$ EC votes
Election → Election Commission	1:1	Each election managed by one commission	Commission formed by Moderator before election starts

Continued on next page

Relationship	Cardinality	Description	Business Rules
Election → Vote	1:N	An Election comprises multiple votes	Votes partitioned by phase (1 or 2); results calculated automatically
Election → ECMember	1:N	An Election produces elected EC members	Winners from Phase 2 assigned to Posts 1–11; Phase 1 winners become EC members
Event → Volunteer	1:N	An Event has registered volunteers	Members sign up; EC assigns specific roles during event
Meeting → MeetingAttendance	1:N	A Meeting records attendance entries	Minimum quorum: 2/3 of EC members; minutes attached to meeting record
Transaction → ExpenditureReport	N:1	Transactions aggregated into quarterly reports	Report generated every 3 months; requires Treasurer and Moderator digital signatures
ConstitutionChange → Vote	1:N	A proposed amendment requires EC votes	Needs $\geq 2/3$ EC approval; then Moderator + Chief Patron + Academic Committee approval
Moderator → ExpenditureReport	1:N	Moderator signs off reports	Must verify all transactions before signing
Moderator → Election Commission	1:N	Moderator forms election commission	Appoints 3 members before election
Chief Patron → Certificate	1:N	Chief Patron approves certificates	Only authority to issue membership certificates
Chief Patron → ConstitutionChange	1:N	Chief Patron approves amendments	Final approval after EC and Moderator votes

KEY CONSTRAINTS AND INTEGRITY RULES

- *Uniqueness*: Each Member has a unique `studentID`; each ECMember has a unique `post` per term (no duplicate posts).
- *Referential Integrity*: All foreign keys (e.g., `voterID`, `candidateID`, `memberID`) must reference valid Member or ECMember records.
- *Status Validation*: Member `status` transitions follow state machine: `Active` → {`Expired`, `Cancelled`}; no reverse transitions allowed.
- *Election Phase Sequencing*: Phase 2 cannot start until Phase 1 results are published; votes are immutable after submission.
- *Financial Audit Trail*: Every Transaction must have `category`, `amount > 0`, `timestamp`, and `createdBy` (Treasurer).
- *Meeting Attendance*: Only current EC members can have attendance entries; absence count auto-increments for no-shows.
- *Certificate Approval*: Certificates can only be issued by users with `role = Chief Patron`; one pending request per member.

- *Term Boundaries:* EC terms are non-overlapping; archived EC records retain historical post assignments for eligibility checks.

ER Diagram: A comprehensive Entity-Relationship diagram illustrating all entities, attributes, primary/foreign keys, and cardinality constraints is provided in **Appendix A** of this document..

DATA RETENTION POLICY

- **Active Records:** Member profiles, current EC data, ongoing elections, and pending transactions retained in primary database.
- **Historical Archives:** Past EC terms, completed elections, and closed financial years moved to archive tables after 1 year; retained for 7 years for audit purposes.
- **Deletion Policy:** Member data anonymized 3 years after graduation unless consent given for alumni engagement.

8 UI and UX Requirements

8.1 Wireframes

The following screens are required as wireframes. See Appendix C for mockups.

- 1 Member Dashboard** — Upcoming events, membership status, notifications feed. (See Figure 3)
- 2 Membership Registration Form** — Multi-step form with checklist sign-off. (See Figure 4)
- 3 Election Portal** — Phase selector, candidate list with eligibility badge, voting interface. (See Figure 5)
- 4 Financial Ledger View** — Tabular ledger with filter by category/date; running balance; export button. (See Figure 6)
- 5 Executive Meeting Scheduler** — Calendar view; meeting creation modal; attendance capture screen. (See Figure 7)
- 6 EC Management Panel** — Org-chart view of current EC; post statuses (filled/vacant); impeachment/resignation triggers. (See Figure 8)

8.2 UX and Accessibility Baseline

- Clear navigation with labelled sidebar and breadcrumb trails.
- All core flows (registration, voting, report submission) are keyboard-accessible.
- Minimum WCAG 2.1 AA colour contrast ratio (4.5:1 for body text).
- Responsive layout for desktop and tablet.
- Error messages are specific and actionable (no generic “Something went wrong”).
- Forms include inline validation and confirmation dialogs for irreversible actions.

9 Non-Functional Requirements

Table 5: Non-Functional Requirements

NFR ID	Category	Requirement	Target / Metric
NFR-CSC-001	Performance	Page load time for dashboard under normal load	≤ 3 seconds at 95th percentile
NFR-CSC-002	Performance	Election vote submission response time	≤ 1 second per vote
NFR-CSC-003	Security	All passwords must be stored using a strong hashing algorithm (bcrypt/Argon2)	Zero plaintext passwords in DB
NFR-CSC-004	Security	All data in transit must use HTTPS/TLS 1.2+	100% encrypted traffic
NFR-CSC-005	Security	Role-based access control enforced on every API endpoint	0 unauthorised access incidents in pen test
NFR-CSC-006	Availability	System uptime during election windows	$\geq 99.5\%$
NFR-CSC-007	Availability	Scheduled maintenance windows communicated 48 hours in advance	100% advance notice
NFR-CSC-008	Usability	New user can complete membership registration without external help	Task completion rate $\geq 90\%$ in usability test
NFR-CSC-009	Usability	System UI must be available in Bangla and English	Language toggle on all pages
NFR-CSC-010	Scalability	System must handle concurrent users during election	≥ 500 concurrent users without degradation

Continued on next page...

NFR ID	Category	Requirement	Target / Metric
NFR-CSC-011	Auditability	All financial transactions must be immutable once saved	No delete option; only adjustment entries allowed
NFR-CSC-012	Reliability	Data backup frequency	Daily automated backup; recovery tested monthly
NFR-CSC-013	Maintainability	Codebase must include unit tests for core business logic	$\geq 80\%$ code coverage

10 Risks and Open Questions

10.1 Risk Register

Table 6: Risk Register

Risk ID	Description	Probability	Impact	Mitigation
RISK-CSC-001	Low student adoption; members prefer manual processes	Medium	High	Onboarding workshops; intuitive UI; support from Moderator
RISK-CSC-002	University network downtime during election	Low	Critical	Offline voting fallback; cached ballot; retry mechanism
RISK-CSC-003	Data breach exposing member PII	Low	Critical	Encryption at rest and transit; regular pen testing; RBAC
RISK-CSC-004	Constitutional ambiguity causing conflicting system rules	Medium	High	Formal review of constitution with Moderator before development

Continued on next page...

Risk ID	Description	Probability	Impact	Mitigation
RISK-CSC-005	Delay in Academic Committee approval for constitution changes	Medium	Medium	Early submission of proposals; parallel informal reviews
RISK-CSC-006	Incomplete financial records from prior EC terms	High	Medium	Data migration plan; import templates for historical data
RISK-CSC-007	Scope creep from EC requests for additional features	Medium	Medium	Strict change-control process; feature backlog managed by Moderator

10.2 Open Questions

- 1 Will the university provide an official email domain for CSEDUSC members, or will personal email addresses be used for authentication?
- 2 What is the authoritative source for “final result published” to trigger automatic membership expiry?
- 3 Should alumni have login access to the system, or is their involvement limited to email/in-person?
- 4 Who is responsible for hosting and maintaining the system server – the department IT team or an external provider?
- 5 How should the system handle elections when a batch has fewer than five willing candidates?
- 6 Is there a budget ceiling per event that the system should enforce, or is Moderator approval sufficient?

11 Testability and Traceability

11.1 Requirement Traceability

Table 7: Requirement Traceability Matrix (partial)

Requirement ID	Story / Use Case	Planned Test ID
FR-CSC-001	US-CSC-001	TC-CSC-001
FR-CSC-002	US-CSC-001	TC-CSC-002
FR-CSC-003	US-CSC-001	TC-CSC-003
FR-CSC-004	US-CSC-008	TC-CSC-004
FR-CSC-005	US-CSC-013, UC-01	TC-CSC-005
FR-CSC-006	US-CSC-013, UC-01	TC-CSC-006
FR-CSC-007	US-CSC-013	TC-CSC-007
FR-CSC-008	US-CSC-008	TC-CSC-008
FR-CSC-009	US-CSC-009	TC-CSC-009
FR-CSC-011	US-CSC-004	TC-CSC-010
FR-CSC-012	US-CSC-005, UC-02	TC-CSC-011
FR-CSC-018	US-CSC-006, UC-03	TC-CSC-012
FR-CSC-019	US-CSC-007, UC-03	TC-CSC-013
FR-CSC-023	US-CSC-014	TC-CSC-014
FR-CSC-024	All user stories	TC-CSC-015

11.2 Initial Manual Test Plan

Table 8: Manual Test Cases

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-CSC-001	Successful member registration	Valid CSEDU student ID exists	1. Open registration page. 2. Fill all fields. 3. Sign checklist. 4. Submit.	Member appears in Active member list; confirmation email received.
TC-CSC-002	Automatic membership expiry on final result	Member is Active; result published flag set	1. Trigger result-published event. 2. Wait for batch job.	Member status changes to “Expired” within 24 hours.
TC-CSC-003	Membership cancellation notice generation	Cancellation initiated by Moderator	1. Moderator initiates cancellation. 2. Chief Patron and President sign.	PDF notice generated with all three signatures; member status is “Cancelled”.
TC-CSC-004	EC formation with all 11 posts	Election results recorded	1. Navigate to EC Management. 2. Assign winners to posts 1–11.	All posts filled; org chart updated; EC members notified.
TC-CSC-005	Phase 1 election voting	Election activated; voter is a batch member	1. Log in. 2. Open Phase 1 ballot. 3. Select up to 5 candidates. 4. Submit vote.	Vote recorded; confirmation shown; cannot vote again.
TC-CSC-006	Phase 2 election with ineligible candidate blocked	Phase 1 complete; candidate has <2 years EC experience for President post	1. Open Phase 2 ballot for President.	Ineligible candidate does not appear; error message displayed if direct access attempted.
TC-CSC-007	Eligibility check for General Secretary candidate	Candidate has 1 year EC experience	1. Attempt to add candidate to General Secretary ballot.	System rejects candidacy with message: “Minimum 2 years EC experience required”.
TC-CSC-008	Impeachment accepted with $\geq 2/3$ vote	EC has 15 members; 10 vote in favour	1. President initiates impeachment. 2. 10 EC members vote yes.	Impeachment accepted; notice generated; member removed from EC.
TC-CSC-009	Impeachment rejected with <2/3 vote	EC has 15 members; 9 vote in favour	1. President initiates impeachment. 2. 9 EC members vote yes.	Impeachment not accepted; member remains in EC; result logged.

Continued on next page

Test ID	Scenario	Preconditions	Steps	Expected Result
TC-CSC-010	Executive meeting scheduled with notification	User is President or General Secretary	1. Create meeting for next week. 2. Save.	Notification sent to all EC members; meeting appears on calendar.
TC-CSC-011	Third consecutive absence flagging	EC member absent from 2 previous meetings	1. Record attendance for third meeting without this member.	Alert sent to President and Moderator; member flagged for potential EC membership cancellation.
TC-CSC-012	Financial ledger entry by Treasurer	User has Treasurer role	1. Open ledger. 2. Add income entry of BDT 5000, category "Donation". 3. Save.	Entry appears in ledger; running balance updates; entry is immutable after saving.
TC-CSC-013	Quarterly expenditure report generation	Three months of transactions recorded	1. Select quarter. 2. Generate report. 3. Treasurer signs. 4. Submit to Moderator.	Report generated correctly; routed to Moderator; archived after Moderator signature.
TC-CSC-014	Certificate of membership request	User is an EC member	1. Navigate to Certificate section. 2. Request certificate.	Request appears in Chief Patron's approval queue; PDF generated after approval; downloadable by member.
TC-CSC-015	RBAC – General member attempts Treasurer action	User is a General Member	1. Log in as General Member. 2. Attempt to access Financial Ledger.	Access denied; HTTP 403 response; redirected to member dashboard.

12 Lab 2 Submission Checklist

- ☒ Problem statement and scope are clear. (*See Sections 2 & 3*)
- ☒ Minimum 12 user stories with acceptance criteria. (*15 stories — see Table 2*)
- ☒ Functional requirements table is complete. (*25 FRs — see Table 1*)
- ☒ Data model and data dictionary are attached. (*See Section 7; ER diagram in Appendix A*)
- ☒ Minimum 5 wireframes or flow screens included. (*6 screens listed — see Section 8.1; mockups in Appendix C*)
- ☒ NFR section is completed. (*13 NFRs — see Table 5*)
- ☒ Risks and open questions are documented. (*See Section 10*)
- ☒ Initial manual test plan (15 cases) is included. (*See Table 8*)

A Entity-Relationship Diagram

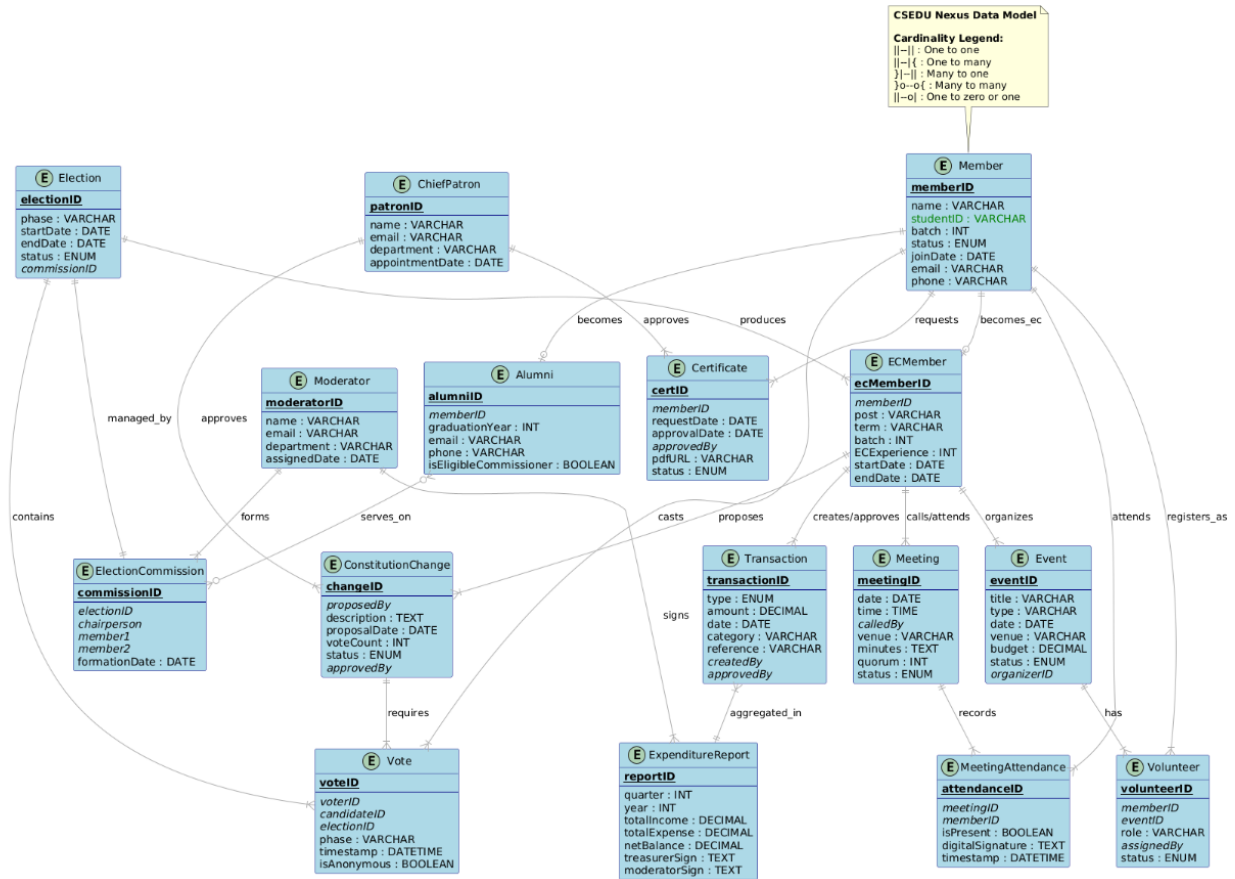


Figure 1: Entity-Relationship Diagram for CSEDU Nexus

Figure 1: Entity-Relationship Diagram for CSEDU Nexus showing 16 entities, their attributes, relationships, and cardinality constraints. Weak entities (Volunteer, MeetingAttendance) depend on their parent entities (Event, Meeting).

B UML Diagram

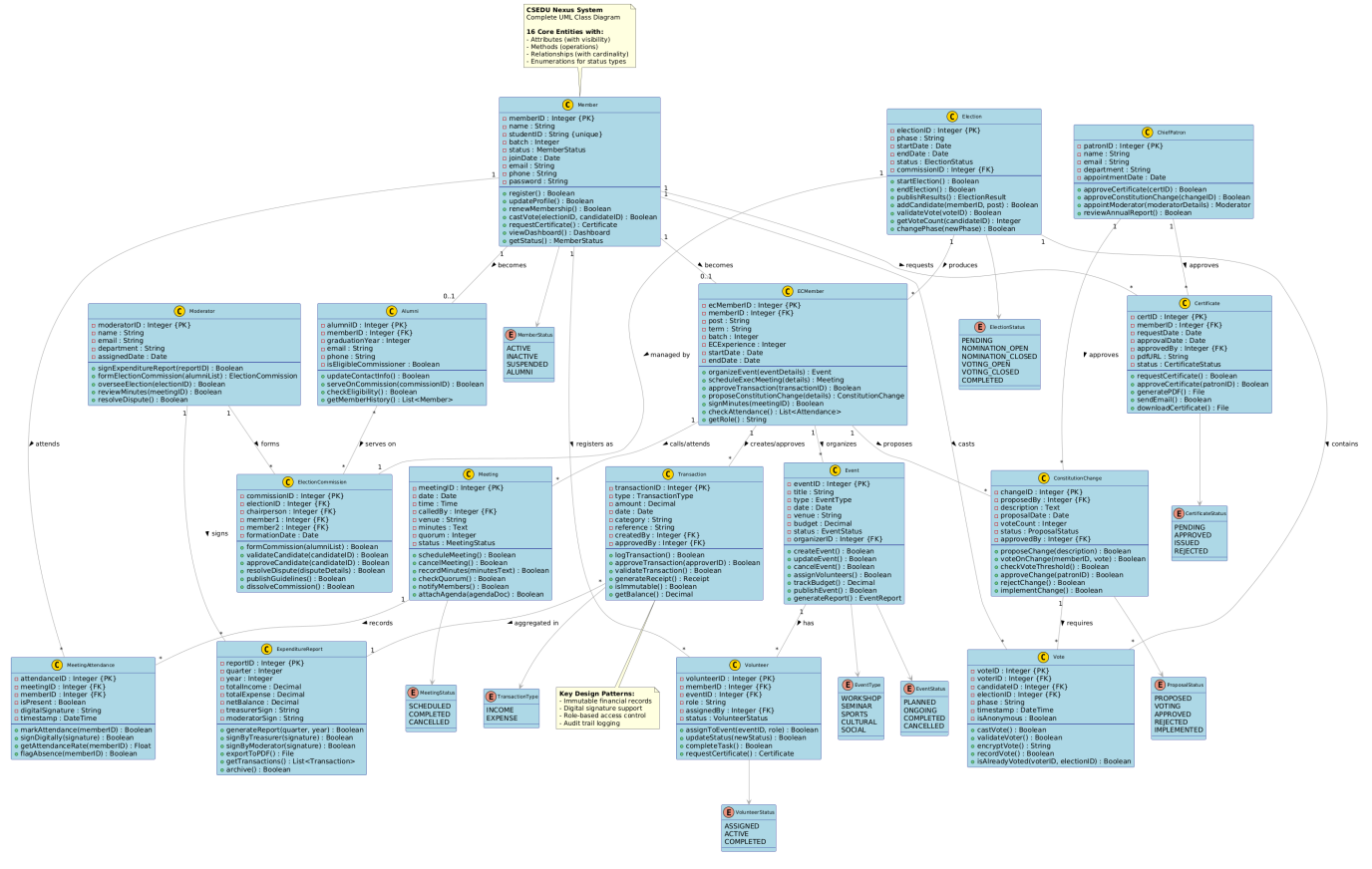


Figure 2: UML Class Diagram of the System

C Wireframes

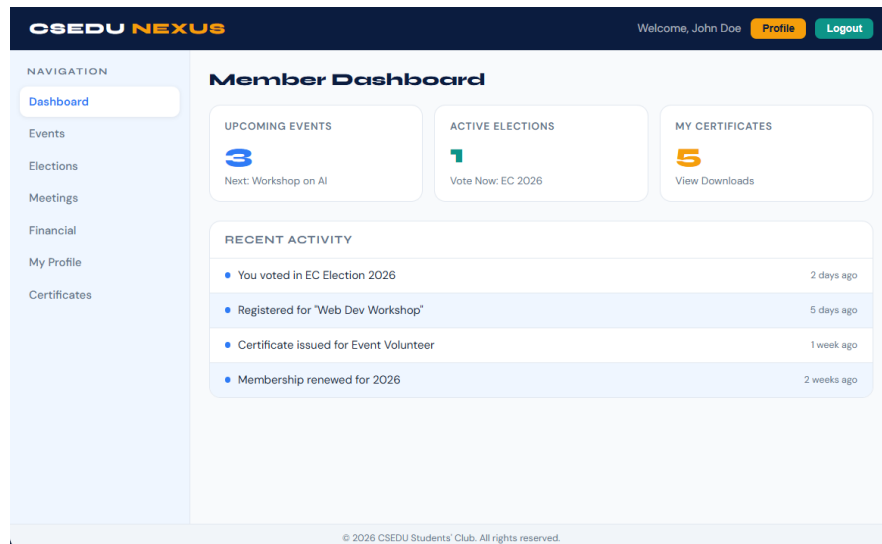


Figure 3: Wireframe 1 - Member Dashboard

This wireframe shows a 'Membership Registration' form for 'CSEDU NEXUS'. The form is titled 'Membership Registration' and includes the following fields: 'FULL NAME' (text input, placeholder: 'Enter your full name'), 'STUDENT ID' (text input, placeholder: 'e.g., 1905001'), 'EMAIL ADDRESS' (text input, placeholder: 'student@cse.du.ac.bd'), 'PHONE NUMBER' (text input, placeholder: '+880 1XXXXXXX'), 'BATCH YEAR' (dropdown menu, placeholder: 'Select year'), and 'MEMBERSHIP TYPE' (dropdown menu, placeholder: 'General / Executive'). At the bottom of the form are two buttons: 'Submit Application' (green) and 'Cancel' (grey). A footnote at the bottom states '* Required fields'.

Figure 4: Wireframe 2 – Membership Registration Form

CSEDU NEXUS
Member: John Doe

Executive Committee Election 2026

Position: President

Active
Ends: March 15, 2026

Your Vote: **Not Cast**
Positions Available: 15
Total Candidates: 3

Alice Rahman
ID: 1905012 · Batch: 2019

Tech & Innovation

Vote for Alice

Bob Khan
ID: 1905023 · Batch: 2019

Member Welfare

Vote for Bob

Carol Akter
ID: 1905034 · Batch: 2019

Financial Trans.

Vote for Carol

Instructions: Select one candidate per position. You can vote for up to 15 positions. Click "Confirm Vote" when done.

Confirm Vote **Cancel**

Figure 5: Wireframe 3 – Election Portal

CSEDU NEXUS – Financial Ledger
Treasurer View

TOTAL BALANCE
BDT 1,25,450

THIS MONTH INCOME
+ BDT 35,200

THIS MONTH EXPENSE
- BDT 18,900

Filter: All Types Date Range
+ Add Entry

DATE	TYPE	CATEGORY	DESCRIPTION	AMOUNT (BDT)	BALANCE
2026-03-01	Income	Donation	Corporate sponsor	+ 50,000	1,25,450
2026-02-28	Expense	Event	Workshop materials	- 8,500	75,450
2026-02-25	Income	Membership	Annual fees (15 members)	+ 15,000	83,950
2026-02-20	Expense	Admin	Office supplies	- 3,200	68,950
2026-02-15	Income	Event	Seminar registration	+ 12,000	72,150

Page 1 of 12
Prev
Next

Generate Report **Export to Excel**

Figure 6: Wireframe 4 – Financial Ledger View

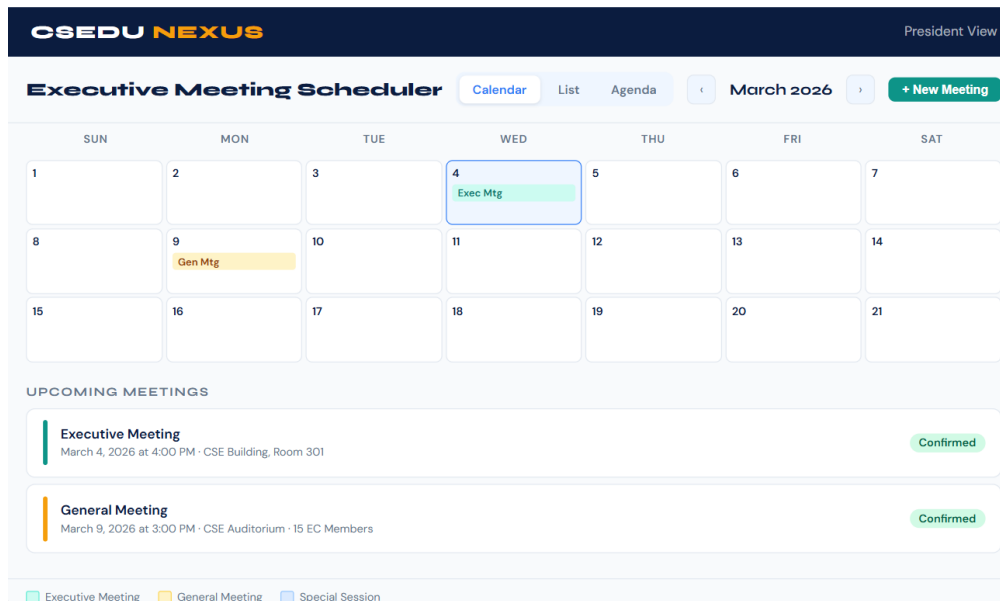


Figure 7: Wireframe 5 – Executive Meeting Scheduler

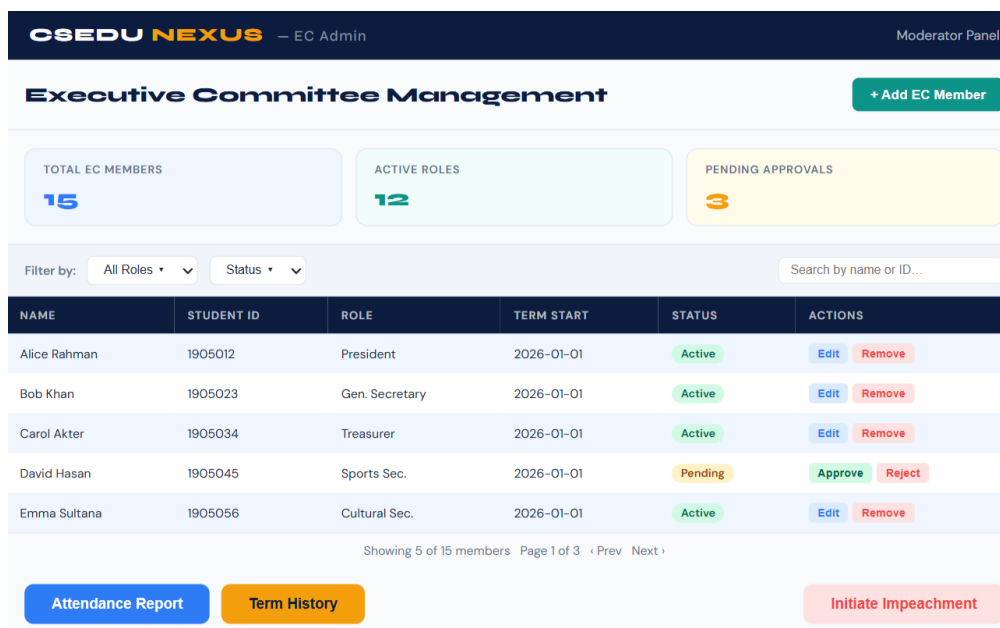


Figure 8: Wireframe 6 – EC Management Panel